

Alessio Ludovico De Santis

Ph.D. Candidate in Astroparticle Physics

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Academic Interests

My research in **multi-messenger astrophysics** focuses on preparing for **next-generation observatories** such as the **Einstein Telescope**. For this I develop **data-analysis software** that combines **decades of data** from gravitational-wave, electromagnetic, and neutrino observatories. My work primarily addresses rare transient such as **core-collapse supernovae** and **binary neutron star mergers**. I study these phenomena through developing methods for **parameter estimation**, **Monte Carlo based forecasts** and identifying unique multi-messenger signatures to probe fundamental physics and extreme astrophysical environments, ultimately pushing the boundaries of multi-messenger astronomy for new insights into the universe's most violent events.

Technical Skills

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| Data Analysis | Bayesian inference, Monte Carlo simulations, regression, parameter estimation, uncertainty quantification, background estimation, and time-series analysis. Analysis and visualisation of scientific data and models from different observatories. |
| Programming | Python (advanced / daily): NumPy, SciPy, pandas, scikit-learn, matplotlib, emcee, and dynesty for analysis, modelling and visualization.
C++ (proficient): Hardware-triggering systems, real-time data, and ray-tracing simulations. |
| Development | Linux/Bash, Git/GitHub, documentation, API integration, performance optimisation, and legacy Python 2 to Python 3 migration. Working knowledge of SQL, JavaScript (plugin, user interfaces), and HTML/CSS. |
| Additional | Geospatial processing with GeoPandas, vector and raster data, coordinate reference systems, and multi-spectral imagery. |

Selected Software & Data Projects

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| MAGGPY | [GitHub] [Paper] Developed and documented a public Python Monte Carlo package connecting binary neutron-star models with gamma-ray burst observations. Optimised it for large parameter searches and use in peer-reviewed research. |
| GBM OSV | [GitHub] Built Python background-estimation tools for the Fermi-GBM detector, used in analyses of GRB 221009A (the BOAT) and ultra-long transients. |
| CAPIBARA | [GitHub] Developed ray-tracing simulations of cryogenic detectors with experimental physicists investigating unexpected sensor behaviour. |
| GWFish | Contributor to an international gravitational-wave forecasting framework; implemented high-frequency corrections and arbitrary frequency- and time-series injections. |
| AQUILA | Small cross-disciplinary project comparing regression approaches on satellite data and spatial embeddings to study socioeconomic indicators. |

Research Experience

- 2022–Present **Doctoral Researcher**, *Gran Sasso Science Institute (GSSI)*, L'Aquila, Italy
- Develop data-analysis frameworks and scientific software for multi-messenger astrophysics, combining gravitational-wave, electromagnetic, and neutrino observations and models.
 - Lead projects from problem formulation through peer-reviewed publication, working with international collaborations. **Publication record includes eight refereed or accepted journal articles and three proceedings, with first- and corresponding-author work.**
- Oct 2025 **Visiting Ph.D. Researcher**, *Niels Bohr Institute*, Copenhagen, Denmark
- Joined a new supernova project, learned its analysis framework, and wrote software to analyse neutrino signals and generate gravitational-wave templates.
- Apr–Sep 2024 **Visiting Ph.D. Researcher**, *Max Planck Institute for Physics*, Munich, Germany
- Worked with experimentalists to develop detector simulations and investigate unexpected low-energy sensor behaviour.

Education

- 2022–Present **Ph.D. in Astroparticle Physics**, *Gran Sasso Science Institute (GSSI)*
Advisors: Marica Branchesi, Samuele Ronchini, Filippo Santoliquido.
- 2022 **M.S. in Particle and Astroparticle Physics**, *University "La Sapienza"*, (*Cum Laude*)
Advisors: Marco Vignati, Angelo Cruciani.
- 2020 **B.S. in Physics**, *University "La Sapienza"*
Advisors: Marco Vignati, Angelo Cruciani.

Communication, Academic Mentorship & Community Engagement

- Presenting Invited seminar speaker at the [Niels Bohr Institute](#) and contributed speaker at international meetings including [COSPAR](#), [THESEUS](#), and the [European Astronomical Society](#).
- Student Supervision Guiding a master's student on leveraging Dark Matter detectors for joint supernova-neutrino detection and mentoring the development of the analysis code.
- Peer Mentorship and Outreach **Outreach:** Speaker/organizer for "Scienza in Bidda" (Sardinia, 2024), tutor for "Space Explorers" (GSSI, 2023–2026).
GSSI Buddies Program: Active mentor for incoming international students, helping them settle into the institute and navigate practical administration in Italy (2025).
GSSI Visitors Center Guide: Preparing interactive scientific experiments and demonstrations, leading educational workshops, and explaining the experiments to visitors (2026).

Languages

- Italian:** Native; lived and completed higher education in Italy, in Rome and L'Aquila.
- English:** Native/bilingual; lived and studied in Canada and use English daily in international research.
- French:** Professional working proficiency; earned a French Baccalauréat after completing middle and high school in French.
- German:** Professional working proficiency; lived in Germany for four years and returned for a six-month research visit at Max Planck in Garching.
- Russian:** Proficient; lived in Russia for four years.
- Lithuanian:** Conversational; lived in Lithuania for four years.